Amdahl's Law Calculations and Analysis

This document contains solutions to speedup problems using Amdahl's Law and includes a graphical representation of the speedup versus the number of processors for varying levels of parallelization.

Problem 1: Speedup Calculation (P = 0.60, N = 10)

Given:

Parallel Portion (P) = 0.60

Serial Portion (S) = 0.40

Number of Processors (N) = 10

Solution:

Using Amdahl's Law:

Speedup = 1 / ((1 - P) + P / N)

Substitute the values:

Speedup = 1 / (0.40 + 0.06) = 2.17

Thus, the speedup is approximately 2.17x.

Problem 2: Speedup Calculation (P = 0.90, N = 10)

Given:

Parallel Portion (P) = 0.90

Serial Portion (S) = 0.10

Number of Processors (N) = 10

Solution:

Speedup = 1 / ((1 - P) + P / N)

Speedup = 1 / (0.10 + 0.09) = 5.26

Thus, the speedup is approximately 5.26x.

Problem 3: Speedup Calculation (P = 0.99, N = 10)

Given:

Parallel Portion (P) = 0.99

Serial Portion (S) = 0.01

Number of Processors (N) = 10

Solution:

Speedup = 1 / ((1 - P) + P / N)

Speedup = 1 / (0.01 + 0.099) = 9.17

Thus, the speedup is approximately 9.17x.

Graph: Speedup vs Number of Processors

The graph below shows the speedup as the number of processors increases for different parallel portions of the code. As predicted by Amdahl's Law, speedup increases rapidly at first but reaches a limit determined by the serial portion of the code.